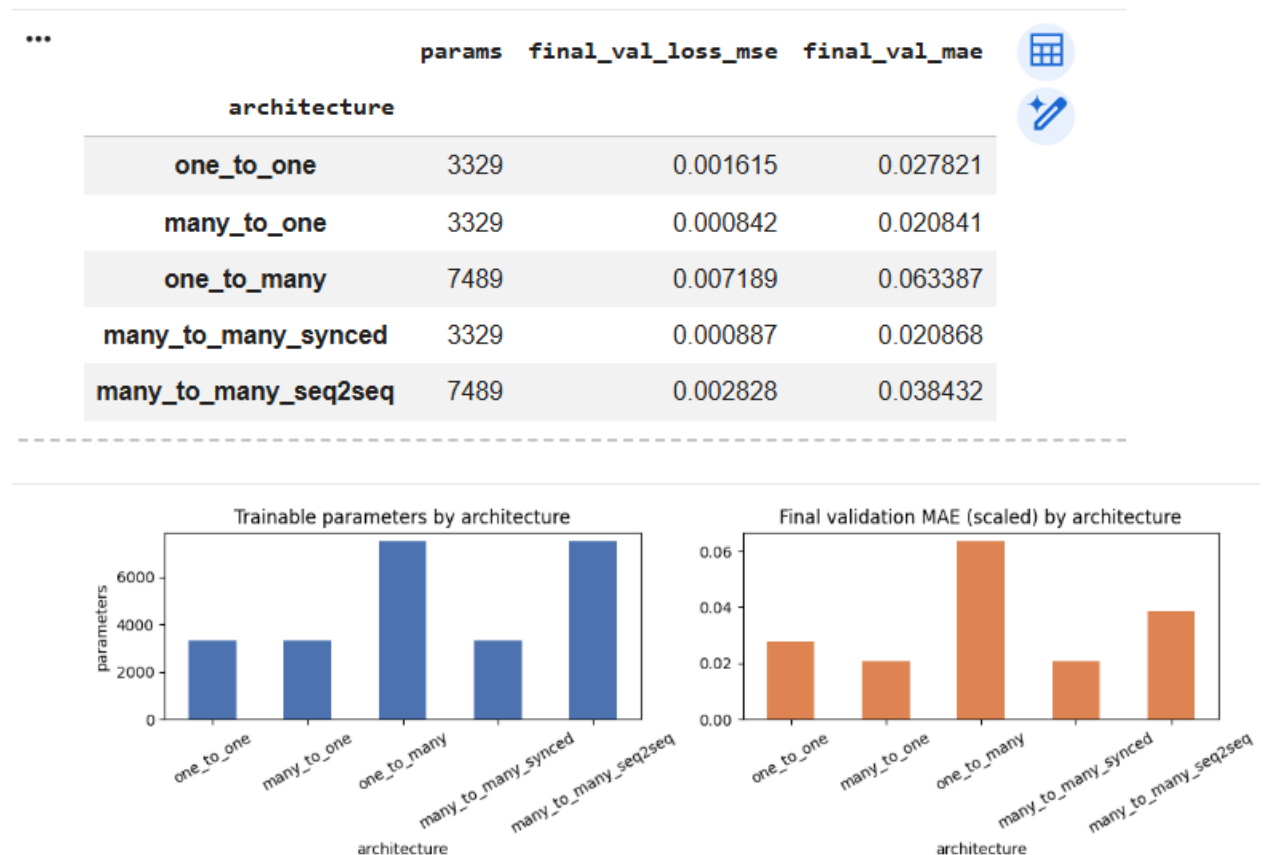


## Activity RNN, LSTM, GRU Hands-on.

1. Based on the notebooks provided. add another layer of RNN, LSTM, GRU. with same number of units/neurons.
  2. Get all metrics MAE, MSE, RMSE,  $r^2$  scores of each. each typology
  3. What typology is best when we add another layer of each type.
- 
4. Advance (challenge), Add another layer of CNN (just 1 layer) before RNN and run for 100 epochs. number of filters = 32, filter/kernel size = 3
  5. Give your conclusion about the performance of item number 4 settings.

## Answers:

## Vanila\_RNN



| TYPOLOGY             | PARAMS | MAE    | MSE     | RMSE   | R2     |
|----------------------|--------|--------|---------|--------|--------|
| one_to_one           | 3329   | 2.3051 | 9.6505  | 3.1065 | 0.8455 |
| many_to_one          | 3329   | 1.7537 | 5.142   | 2.2676 | 0.9179 |
| one_to_many          | 7489   | 4.7692 | 41.9647 | 6.478  | 0.3288 |
| many_to_many_synced  | 3329   | 1.6217 | 4.8441  | 2.2009 | 0.9227 |
| many_to_many_seq2seq | 7489   | 3.452  | 19.3456 | 4.3984 | 0.6914 |

LSTM

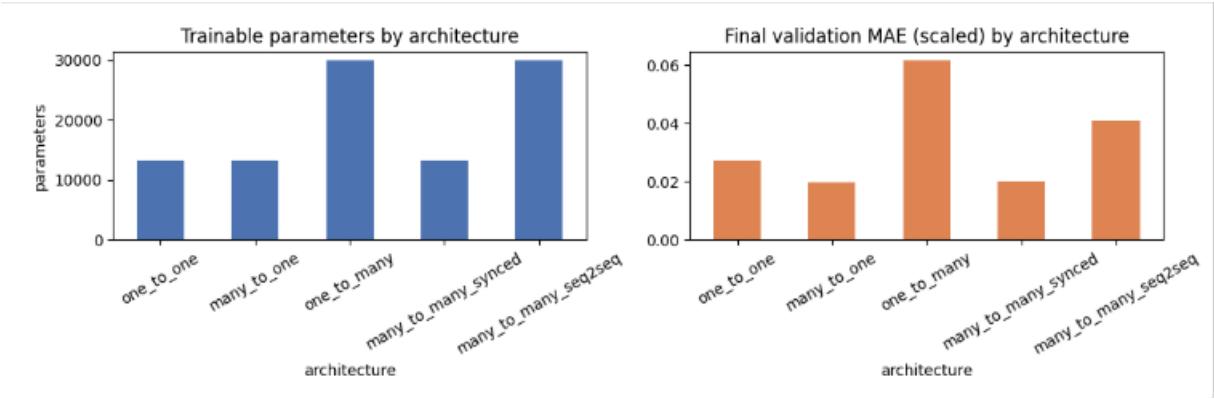
params

final\_val\_loss\_mse

final\_val\_mae

architecture

|                      |       |          |          |
|----------------------|-------|----------|----------|
| one_to_one           | 13217 | 0.001582 | 0.027320 |
| many_to_one          | 13217 | 0.000767 | 0.019472 |
| one_to_many          | 29857 | 0.006762 | 0.061343 |
| many_to_many_synced  | 13217 | 0.000840 | 0.020126 |
| many_to_many_seq2seq | 29857 | 0.003325 | 0.040704 |



| TPOLOGY              | PARAMS | MAE    | MSE     | RMSE   | R2     |
|----------------------|--------|--------|---------|--------|--------|
| one_to_one           | 13217  | 2.0159 | 8.3166  | 2.8838 | 0.8669 |
| many_to_one          | 13217  | 1.5417 | 4.267   | 2.0657 | 0.9319 |
| one_to_many          | 29857  | 4.6851 | 42.5816 | 6.5255 | 0.319  |
| many_to_many_synced  | 13217  | 1.583  | 4.7148  | 2.1714 | 0.9248 |
| many_to_many_seq2seq | 29857  | 3.4052 | 20.4038 | 4.5171 | 0.6745 |

GRU

params

final\_val\_loss\_mse

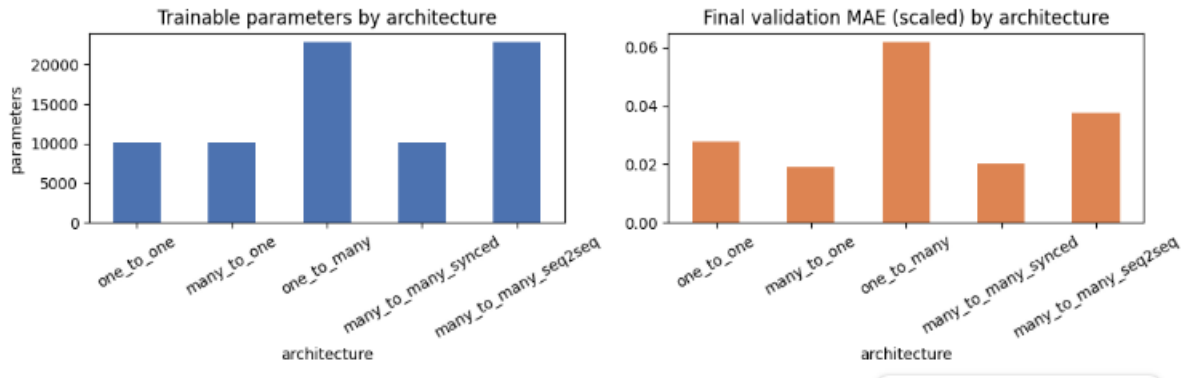
final\_val\_mae

architecture

|                      |       |          |          |
|----------------------|-------|----------|----------|
| one_to_one           | 10113 | 0.001629 | 0.027681 |
| many_to_one          | 10113 | 0.000761 | 0.019140 |
| one_to_many          | 22785 | 0.006877 | 0.061676 |
| many_to_many_synced  | 10113 | 0.000850 | 0.020081 |
| many_to_many_seq2seq | 22785 | 0.002821 | 0.037639 |



| TPOLOGY              | PARAMS | MAE    | MSE     | RMSE   | R2     |
|----------------------|--------|--------|---------|--------|--------|
| one_to_one           | 10113  | 2.1814 | 9.1803  | 3.0299 | 0.853  |
| many_to_one          | 10113  | 1.4753 | 3.9656  | 1.9914 | 0.9367 |
| one_to_many          | 22785  | 4.683  | 41.9763 | 6.4789 | 0.3286 |
| many_to_many_synced  | 10113  | 1.5856 | 4.7713  | 2.1843 | 0.9239 |
| many_to_many_seq2seq | 22785  | 3.1152 | 16.9482 | 4.1168 | 0.7296 |



### Item 1: Adding another layer

Trainable parameters after stacking:

| Typology                          | SimpleRNN | LSTM   | GRU    |
|-----------------------------------|-----------|--------|--------|
| one_to_one / many_to_one / synced | 3,329     | 13,217 | 10,113 |
| one_to_many / seq2seq             | 7,489     | 29,857 | 22,785 |

LSTM has about 4 times the parameters of SimpleRNN (4 sets of gate weights). GRU has about 3 times (3 sets).

### Item 2: Test-set metrics (2 layers, real Kaggle data, MAE and RMSE in EUR/MWh)

| Network   | Typology             | MAE    | MSE     | RMSE   | R <sup>2</sup> |
|-----------|----------------------|--------|---------|--------|----------------|
| SimpleRNN | one_to_one           | 2.3051 | 9.6505  | 3.1065 | 0.8455         |
| SimpleRNN | many_to_one          | 1.7537 | 5.1420  | 2.2676 | 0.9179         |
| SimpleRNN | one_to_many          | 4.7692 | 41.9647 | 6.4780 | 0.3288         |
| SimpleRNN | many_to_many_synced  | 1.6217 | 4.8441  | 2.2009 | 0.9227         |
| SimpleRNN | many_to_many_seq2seq | 3.4520 | 19.3456 | 4.3984 | 0.6914         |

| Network | Typology             | MAE    | MSE     | RMSE   | R <sup>2</sup> |
|---------|----------------------|--------|---------|--------|----------------|
| LSTM    | one_to_one           | 2.0159 | 8.3166  | 2.8838 | 0.8669         |
| LSTM    | many_to_one          | 1.5417 | 4.2670  | 2.0657 | 0.9319         |
| LSTM    | one_to_many          | 4.6851 | 42.5816 | 6.5255 | 0.3190         |
| LSTM    | many_to_many_synced  | 1.5830 | 4.7148  | 2.1714 | 0.9248         |
| LSTM    | many_to_many_seq2seq | 3.4052 | 20.4038 | 4.5171 | 0.6745         |
| GRU     | one_to_one           | 2.1814 | 9.1803  | 3.0299 | 0.8530         |
| GRU     | many_to_one          | 1.4753 | 3.9656  | 1.9914 | 0.9367         |
| GRU     | one_to_many          | 4.6830 | 41.9763 | 6.4789 | 0.3286         |
| GRU     | many_to_many_synced  | 1.5856 | 4.7713  | 2.1843 | 0.9239         |
| GRU     | many_to_many_seq2seq | 3.1152 | 16.9482 | 4.1168 | 0.7296         |

### Item 3: Which typology is best with the extra layer?

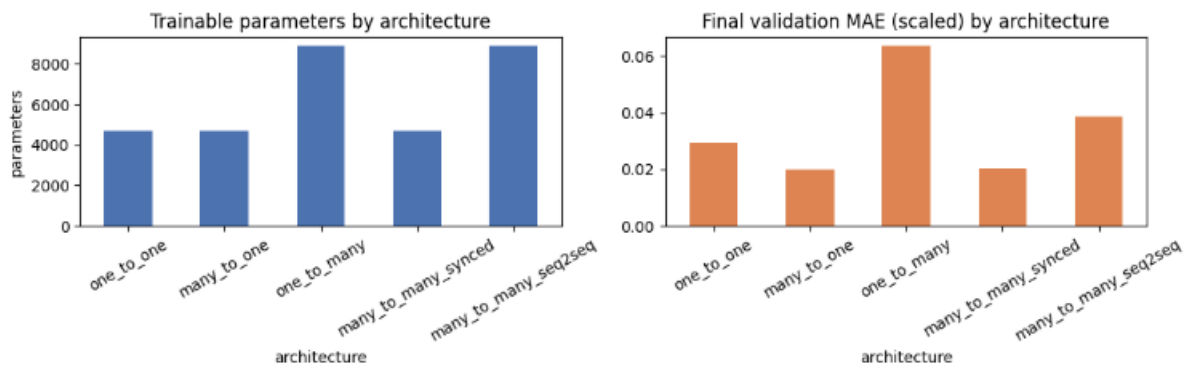
Many-to-one is the best typology. It has the highest R<sup>2</sup> for all three network types, and the best single model is the stacked GRU many-to-one (MAE 1.48 EUR/MWh, RMSE 1.99, R<sup>2</sup> 0.937).

### Which type did best overall?

GRU has the highest average R<sup>2</sup> (0.754), then LSTM (0.743), then SimpleRNN (0.741). GRU wins many-to-one and seq2seq, and LSTM wins one-to-one and synced by small margins.

## ANSWER TO CHALLENGE QUESTION:

|                      | params | MAE    | MSE     | RMSE   | R2     |
|----------------------|--------|--------|---------|--------|--------|
| typology             |        |        |         |        |        |
| one_to_one           | 4705   | 2.5837 | 11.5732 | 3.4019 | 0.8147 |
| many_to_one          | 4705   | 1.5608 | 4.4022  | 2.0981 | 0.9297 |
| one_to_many          | 8865   | 4.5274 | 38.1874 | 6.1796 | 0.3892 |
| many_to_many_synced  | 4705   | 1.5938 | 4.7703  | 2.1841 | 0.9239 |
| many_to_many_seq2seq | 8865   | 3.4042 | 18.7594 | 4.3312 | 0.7008 |



## Conclusions:

1. The CNN helped 4 of 5 typologies. The biggest gain is in many-to-one: MAE dropped from 1.75 to 1.56 EUR/MWh. The CNN picks up short 3-hour patterns (like a morning demand ramp or solar coming online), so the SimpleRNN gets cleaner signals to work with.
2. SimpleRNN + CNN many-to-one (MAE 1.56) now nearly matches the stacked LSTM (1.54), with about 3 times fewer parameters (4,705 vs 13,217). The CNN takes over part of the feature-extraction job, which lets a simple cell perform like a more complex one.
3. One-to-one got worse. With only 1 timestep, a 3-hour kernel has nothing to scan across. Causal padding fills the missing 2 hours with zeros, so the CNN just adds extra weights without finding any real pattern. It's a good example of a layer that doesn't fit the input shape.
4. One-to-many improved the most in  $R^2$  (0.33 to 0.39), though it's still the weakest typology overall.